Dynamic Contracts and Corruption: A Recursive Macroeconomic Approach

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Abstract

We develop a recursive dynamic contracting model in which a self-interested government designs a tax-and-transfer scheme for risk-averse citizens who face idiosyncratic productivity shocks and can divert effort into an untaxed informal sector. The government is subject to two-sided physical limited commitment: it cannot commit not to expropriate the public capital stock it installs for the next period, and citizens can credibly threaten revolution if promised utility falls below a state-dependent outside option that depends on current capital and on the state's enforcement capacity $\kappa \geq 0$. The state variables are the promised continuation utility to citizens and the stock of public infrastructure capital.

The framework yields three results. First, we derive two distinct political-economy wedges on the return to public capital: a *revolution tax* that arises because higher current capital tightens citizens' participation constraints, and an *expropriation tax* that arises because higher future capital tightens the government's no-deviation constraint. Both wedges are strictly positive whenever corruption is positive, generating chronic under-investment. Second, the corruption multiplier enters the per-type labor wedge through both the resource shadow price (a uniform scaling consistent with Acemoglu et al. [2010]) and the incentive multiplier (a non-separable correction). Numerical experiments show that the uniform scaling dominates: the labor distortion is increasing in corruption across nearly the entire plausible parameter space, with narrow regions in which the corruption-disciplining effect dominates. Third, the inverse Euler equation survives in expectation despite the political-economy frictions, providing a clean benchmark for quantitative implementation. The framework nests dynamic Mirrlees taxation and limited-commitment risk-sharing as special cases.

Keywords: Corruption, dynamic contracts, limited commitment, public investment, informal sector, optimal taxation, political economy.
JEL Codes: D82, D86, E62, H21, H54, O17, O43.

1 Introduction

Corruption, sizable informal sectors, and chronically low public investment are jointly determined features of many developing economies. Cross-country data show systematic negative correlations between corruption perceptions and both public capital stocks and formal employment shares

(Mauro 1995; Kaufmann et al. 2011; Medina and Schneider 2018). Yet most macroeconomic models address these phenomena in isolation. Political-economy models of limited commitment typically abstract from private information and informal production (Thomas and Worrall 1994; Aguiar and Amador 2011); dynamic Mirrlees models of optimal taxation assume a benevolent planner (Golosov et al. 2003; Albanesi and Sleet 2006; Kocherlakota 2010); and informal-sector models with rich heterogeneity also maintain a benevolent government (Doligalski and Rojas 2023; da Costa and Lobel 2024). Acemoglu et al. (2010) represent the closest existing point of contact: they study dynamic Mirrlees taxation operated by a self-interested politician disciplined by replacement and prove a separation result whereby political-economy distortions remain aggregate while per-type Mirrleesian wedges are unaffected.

This paper develops a unified framework that embeds corruption inside a recursive dynamic contract between a self-interested government and risk-averse citizens. The government is subject to two-sided physical limited commitment: it can expropriate the public capital stock it installs for the next period, and citizens can credibly threaten revolution if promised utility falls below a state-dependent outside option. The threshold for revolution depends both on current capital—richer economies provide a more attractive post-revolution endowment—and on an enforcement parameter $\kappa \geq 0$ that captures the state's capacity to monitor, repress, or absorb discontent through public goods. Citizens face idiosyncratic productivity shocks and can divert effort to an untaxed informal sector, generating a Mirrleesian information friction.

The contribution is threefold. **First**, we derive two conceptually and mathematically distinct political-economy wedges on the return to public capital. The *revolution tax* arises because higher current capital raises citizens' outside option, tightening their participation constraints. The *expropriation tax* arises because higher future capital tightens the government's no-deviation constraint tomorrow, requiring lower investment today to remain self-enforcing. The two wedges operate through different multipliers (citizen participation vs. government incentive) and cannot be collapsed into a single term by a change of state variable. Both are strictly positive whenever the equilibrium features positive corruption. Together they generate chronic under-investment in public infrastructure relative to the first-best and the constrained-efficient benchmark.

Second, we examine how the corruption multiplier interacts with the per-type Mirrlees wedge on labor supply. The intratemporal optimality condition yields a labor distortion that decomposes additively into an aggregate corruption component (uniform scaling of the resource shadow price) and a type-specific Mirrleesian informational component (scaled by the same shadow price). Strict separation in the Acemoglu-Golosov-Tsyvinski (2010) sense fails because the corruption multiplier enters both terms and the equilibrium incentive multiplier responds endogenously. However, numerical analysis across the plausible parameter space shows that the aggregate corruption distortion dominates: the labor distortion is increasing in corruption almost everywhere. The parameter region where the indirect channel reverses the sign is narrow and requires specific combinations of high informal-labor elasticity and slack participation. The resulting picture is closer to "approximate separation with small non-separable corrections" than to a sharp breakdown.

Third, we show that the inverse Euler equation survives in expectation despite the political-economy frictions, exactly as in the benevolent-planner benchmark. The cancellation between the government's no-deviation multiplier in the consumption and continuation-utility first-order conditions is exact, provided two scope conditions hold: the government's deviation value depends only on aggregate quantities (including the capital it can seize), and the participation multipliers enter the consumption and continuation-utility conditions symmetrically. Both conditions are satisfied

in our environment. The result provides a clean benchmark for quantitative implementation.

The framework is designed for quantitative work. The triple of primitives (ϕ, ψ, κ)—loosely interpretable as corruption intensity, informal-sector strength, and enforcement capacity—maps to observables (corruption perception indices, informal-employment shares, and indicators of state capacity). The model nests both the constrained-efficient Mirrlees economy ($\phi = 0$, $\xi_i = 0$) and standard limited-commitment risk-sharing models without private information (θ_i degenerate). We sketch the computational strategy and discuss its non-convexity issues but defer full quantitative implementation and calibration to a companion paper.

Related literature. The paper sits at the intersection of four literatures. The first is dynamic optimal taxation under private information (Mirrlees 1971; Golosov et al. 2003; Kocherlakota 2010; Albanesi and Sleet 2006; Farhi and Werning 2013), which we extend by allowing the planner to be self-interested and subject to limited commitment. The second is the literature on self-enforcing contracts and limited commitment (Thomas and Worrall 1988, 1994; Kocherlakota 1996; Kehoe and Perri 2002; Ligon et al. 2002; Marcet and Marimon 2019), which we extend by adding private information and an informal sector. The third is the political economy of dynamic fiscal policy (Acemoglu and Robinson 2006; Acemoglu et al. 2008a,b, 2010, 2011; Farhi et al. 2012; Yared 2010), where the closest comparator is Acemoglu et al. (2010). They develop a dynamic Mirrlees model with a self-interested politician disciplined by replacement and prove a separation result; in our environment the politician is disciplined by physical expropriation and revolution rather than replacement, and the per-type Mirrlees wedge is approximately—but not exactly—separable from the corruption multiplier. Aguiar and Amador (2011) study growth under expropriation risk with capital and aggregate political frictions; we share their physical-capital deviation structure but add idiosyncratic citizen productivity shocks, private information, and an informal sector. The fourth literature is on the informal sector and optimal taxation (Loayza 1996; Schneider 2005; Bosch and Esteban-Pretel 2012; Doligalski and Rojas 2023; da Costa and Lobel 2024); relative to that work, we replace the benevolent planner with a self-interested government and embed the informal sector inside a recursive contract with two-sided limited commitment.

The remainder of the paper is organized as follows. Section 2 sets up the environment. Section 3 states the recursive problem and discusses existence and the revolutionary breakdown boundary. Section 4 derives the first-order conditions. Section 5 characterizes the two political wedges on public capital. Section 6 analyzes the labor wedge and its relation to AGT separation. Section 7 discusses equilibrium and comparative statics. Section 8 concludes. Appendices contain proofs, computational details, and replication instructions.

2 Environment

Time is discrete and infinite, indexed by $t = 0, 1, 2, \ldots$. The economy is populated by a continuum of citizens of unit measure and a single self-interested government.

2.1 Preferences and Technology

Citizens have additively separable preferences over formal consumption c and informal time l:

$$U(c, l) = u(c) + v(l) = \frac{c^{1-\sigma}}{1-\sigma} + \psi \frac{l^{1-\gamma}}{1-\gamma}, \tag{1}$$

with $\sigma > 0$ the coefficient of relative risk aversion, $\gamma > 0$ the inverse elasticity of informal labor supply, and $\psi > 0$ scaling preferences for informal activity. Each citizen is endowed with one unit of time. Formal labor supply is $e_i = 1 - l_i$. The case $\sigma = 1$ corresponds to logarithmic utility in consumption.

Each citizen draws an i.i.d. idiosyncratic productivity shock $\theta_i \in \Theta = \{\theta_1 < \theta_2 < \cdots < \theta_N\}$ with probability $\pi_i > 0$, $\sum_i \pi_i = 1$. By the law of large numbers the cross-sectional distribution coincides with $\{\pi_i\}$. Output in the formal sector uses public infrastructure capital K as input through the Cobb–Douglas technology

$$y_i = F(K, \theta_i, 1 - l_i) = K^{\alpha}[\theta_i(1 - l_i)]^{1-\alpha}, \quad \alpha \in (0, 1). \tag{2}$$

We denote $F_e(K, \theta, 1 - l) = (1 - \alpha)K^{\alpha}\theta^{1-\alpha}(1 - l)^{-\alpha} > 0$, the marginal product of formal labor. Public capital depreciates at rate $\delta \in (0, 1)$.

The government observes formal output y_i but not the productivity draw θ_i or the informal time choice l_i. Any feasible contract must therefore be incentive-compatible with respect to type reporting and labor allocation.

[Single Crossing] The utility function and production technology satisfy

$$\frac{\partial}{\partial \theta}\left(\frac{v'(l)}{u'(c)F_e(K, \theta, 1 - l)}\right) < 0. \tag{3}$$

Under Assumption 1, only local downward incentive constraints bind at the optimum. Appendix B verifies that this holds for the parameterization in (1)–(2).

2.2 Information and Commitment Frictions

The contract is subject to two commitment frictions, both physical rather than reputational.

Government expropriation. The government cannot commit not to expropriate. If it deviates it seizes the undepreciated capital stock it has installed for the next period, $(1-\delta)K'$, as a one-time payoff, after which it loses power forever and receives zero continuation value:

$$J^{\text{dev}}(K') = (1 - \delta)K'. \tag{4}$$

(The deviation payoff is evaluated at the capital the government itself chose to install, consistent with the timing below.) The deviation is politically terminal: the polity reverts to autarky and the government extracts no further rents.

Revolution threat. Citizens can credibly trigger revolution if their continuation utility falls below the value of revolution. Revolution destroys a fraction $\tau \in (0, 1)$ of the capital stock; thereafter citizens operate a constrained-efficient decentralized economy. We parameterize the revolution value as

$$V^{\text{rev}}(K, \kappa) = [1 - h(K, \kappa)] \cdot V^{\text{eff}}\big(K(1 - \tau)\big), \tag{5}$$

where $V^{\text{eff}}(\cdot)$ is the value function of a benevolent constrained-efficient planner operating the post-revolution economy with capital $K(1-\tau)$, and $h(K, \kappa)$ is a non-negative dampening function with $h(K, 0) = 0$, $\partial h/\partial \kappa > 0$, and $h(K, \kappa) \in [0, 1)$ for all relevant K. Higher κ corresponds to greater state capacity to monitor, repress, or otherwise reduce the appeal of revolution at any given K.

Throughout we assume $V^{\text{rev}}(K, \kappa)$ is increasing in K, decreasing in κ, and continuously differentiable. (The monotonicity of the revolutionary threshold $\underline{V}(K, \kappa)$ is characterized numerically in Section 3.1 rather than assumed a priori.)

Stage game and timing. Within each period the timing is: (i) the government announces an allocation $\{c_i, l_i, W_i\}_{i=1}^N$ together with stolen rents S and next-period capital K'; (ii) citizens decide whether to revolt; (iii) if no revolution, the government decides whether to expropriate.

The terminal payoffs are $V^{\text{rev}}(K, \kappa)$ for revolution, $J^{\text{dev}}(K')$ for expropriation, and the recursive value $J(\bar{W}, K')$ for continuation. Figure 1 summarizes the structure.

Figure 1 here
Stage game and terminal payoffs. The government moves first; citizens then choose whether to revolt; conditional on no revolution, the government chooses whether to expropriate. The continuation value $J(\bar{W}, K')$ corresponds to a self-enforcing path of the recursive contract.

Figure 1: Stage game and terminal payoffs.

This timing makes the expropriation tax on *future* capital economically natural: choosing higher K' directly tightens the government's own no-deviation constraint tomorrow because the deviation payoff rises with K'.

(The interpretation of κ accommodates both the coercive dimension of state capacity emphasized by Besley and Persson 2011 and the cohesion-and-redistribution dimension emphasized by Acemoglu and Robinson 2006. We adopt this reduced-form representation to keep $V^{\text{rev}}(K, \kappa)$ tractable as an exogenous boundary condition; a fully microfounded $h(\cdot)$ would solve a coordination subgame as in Acemoglu et al. 2004, which we leave for future work.)

3 The Recursive Problem

We adopt the Spear and Srivastava (1987) promised-utility formulation. The aggregate state is $\Omega = (V, K)$, where V is the promised continuation utility to a representative citizen and K is the public capital stock. By the i.i.d. assumption on θ_i and the law of large numbers, the cross-sectional distribution of types is deterministic, and the recursive formulation collapses to a single aggregate promised-utility state.

The government's Bellman equation is

$$J(V, K) = \max_{\{c_i, l_i, W_i\}_{i=1}^N, S, K'} S + \beta J(\bar{W}, K') \tag{6}$$

subject to the constraints (with associated multipliers in brackets):

$$\sum_{i=1}^{N} \pi_i \big(u(c_i) + v(l_i) + \beta W_i\big) = V, \qquad [\lambda] \tag{PK}$$

$$\sum_{i=1}^{N} \pi_i c_i + K' + S \le \sum_{i=1}^{N} \pi_i F(K, \theta_i, 1 - l_i) + (1-\delta)K, \qquad [\rho] \tag{RC}$$

$$u(c_i) + v(l_i) + \beta W_i \ge u(c_{i-1}) + v(\hat{l}_{i,i-1}) + \beta W_{i-1}, \quad i = 2, \dots, N, \qquad [\mu_i] \tag{IC}$$

$$u(c_i) + v(l_i) + \beta W_i \ge V^{\text{rev}}(K, \kappa), \quad \forall i, \qquad [\xi_i] \tag{PC}$$

$$S + \beta J(\bar{W}, K') \ge J^{\text{dev}}(K'), \qquad [\phi] \tag{GIC}$$

where $\bar{W} = \sum_i \pi_i W_i$ is the average promised continuation, and $\hat{l}_{i,i-1}$ is the informal time required for type i to mimic type $i-1$'s formal output:

$$F(K, \theta_i, 1 - \hat{l}_{i,i-1}) = F(K, \theta_{i-1}, 1 - l_{i-1}) \quad \Longleftrightarrow \quad 1 - \hat{l}_{i,i-1} = \frac{\theta_{i-1}}{\theta_i}(1 - l_{i-1}). \tag{7}$$

Because $\theta_i > \theta_{i-1}$ implies $\hat{l}_{i,i-1} > l_{i-1}$, mimicking down is costly in informal leisure forgone. We adopt the convention $\mu_1 = \mu_{N+1} = 0$.

[Feasible allocation] An allocation $\{c_i, l_i, W_i, K', S\}(V, K)$ is *feasible* if it satisfies (PK)–(GIC) for all (V, K) in the domain of J.

3.1 Domain Restriction and Revolutionary Breakdown

For each K there exists a state-dependent threshold $\underline{V}(K, \kappa)$ below which (PC) and (GIC) cannot be jointly satisfied; the domain of J is therefore restricted to $\{(V, K) : V \ge \underline{V}(K, \kappa)\}$. When V approaches this threshold from above, no self-enforcing contract exists: revolution becomes the equilibrium outcome.

Unlike the monotonicity claim in earlier drafts, we do not impose that $\underline{V}(K, \kappa)$ is increasing in K. Higher current capital expands the resource base and inside-contract surplus, which tends to *lower* the minimum sustainable promise, while simultaneously raising the outside option $V^{\text{rev}}(K, \kappa)$. The net effect is theoretically ambiguous and is characterized numerically in the computational appendix (Appendix H). We simply maintain that $\underline{V}(K, \kappa)$ is continuous and that the feasible set is non-empty for sufficiently high V or low K.

3.2 Existence and Concavity

[Existence] Let the primitives satisfy $\beta(1-\delta) < 1$, u and v strictly concave, F strictly concave in K and in formal labor $e = 1 - l$, and $V^{\text{rev}}(K, \kappa)$ continuous and increasing in K. There exists a unique bounded continuous function $J(V, K)$ satisfying the Bellman equation (6), and the optimal policy correspondence is non-empty, compact-valued, and upper hemicontinuous on the domain.

The proof (Appendix A) applies the Contraction Mapping Theorem to the Bellman operator on the space of bounded continuous functions, using compactness of the feasible set. Because the (GIC) renders the constraint set in $(K', \bar{W})$ non-convex in general (Cole and Kubler 2012), we

do not claim global concavity of J in K. Instead, we maintain local differentiability at interior optima (justified by lottery-extension arguments or by working directly with the dual recursive formulation of Marcet and Marimon 2019, which treats multipliers as state variables and avoids the concavity question entirely). This is sufficient for the first-order conditions derived in Section 4 and for the quantitative implementation in Appendix H.

4 First-Order Conditions

The Lagrangian for problem (6)–(GIC) is standard and is given in Appendix C. The four economically informative first-order conditions are presented below; full derivations are deferred to the appendix.

4.1 Stolen Rents and the Corruption Wedge

The first-order condition with respect to S yields

$$1 - \rho + \phi = 0 \quad \Longleftrightarrow \quad \rho = 1 + \phi. \tag{8}$$

[The corruption wedge] The shadow price of resources ρ is strictly inflated above unity whenever (GIC) binds ($\phi > 0$). The self-interested government faces an effective marginal cost of public funds equal to $\rho = 1 + \phi$ rather than 1, because every unit of resources allocated to the contract today tightens its own no-deviation constraint relative to the alternative of stealing the installed capital K'. Equation (8) is the closed-form expression of this corruption wedge and propagates through every other first-order condition.

4.2 Consumption and the Modified Risk-Sharing Rule

The first-order condition for c_i is

$$\frac{1}{u'(c_i)} = \frac{\lambda}{\rho} + \frac{\xi_i}{\rho \pi_i} + \frac{\mu_i - \mu_{i+1}}{\rho \pi_i}. \tag{9}$$

This is the modified risk-sharing rule. Because $\rho \geq 1$, with equality if and only if $\phi = 0$, every type's consumption is strictly compressed relative to the benevolent-planner benchmark whenever the government's no-deviation constraint binds. Binding participation constraints ($\xi_i > 0$) further shift consumption toward types whose outside option is binding—typically the lowest types when V is close to the breakdown threshold—while binding incentive constraints ($\mu_i > 0$) preserve the standard Mirrleesian downward distortion of consumption for low types.

4.3 Continuation Utility and the Inverse Euler Equation

The first-order condition for W_i, combined with (9) and the fact that the (GIC) multiplier ϕ enters the continuation-utility FOC only through $\bar{W}$, yields the *inverse Euler equation in expectation*:

$$-J_V(\bar{W}, K') = \sum_{i=1}^{N} \pi_i \frac{1}{u'(c_i)} \qquad \text{for all } (V, K) \text{ in the interior of the domain.} \tag{10}$$

[Inverse Euler] The inverse Euler relation (10) holds at any interior optimum of (6) provided (i) the government's deviation value J^{dev} depends only on aggregate quantities (including the capital it can seize), and (ii) the participation multipliers ξ_i enter the consumption and continuation-utility FOCs symmetrically.

The cancellation between ϕ in the consumption FOC (via $\rho = 1 + \phi$) and ϕ in the continuation FOC is exact; the participation multiplier ξ_i enters identically in the numerators of both $1/u'(c_i)$ and $-J_V$ and cancels under condition (ii). Both conditions hold in our environment: $J^{\text{dev}}(K') = (1-\delta)K'$ depends only on aggregate capital, and (PC) enters both consumption and continuation FOCs through the same Lagrangian term. The proof is in Appendix D.

The result is striking because it shows that the political-economy frictions, while distorting the *level* of consumption, do not destroy the dynamic risk-sharing structure that characterizes constrained-efficient allocations under private information. This provides a clean benchmark for quantitative implementation: an econometrician can recover the distorted shadow price $\rho = 1 + \phi$ from observed consumption growth and capital returns, with the expropriation and revolution taxes appearing as testable residuals.

4.4 The Labor Wedge

Substituting (9) into the first-order condition for l_i and using the correct implicit-function relation

$$\frac{\partial \hat{l}_{i+1,i}}{\partial l_i} = \frac{F_e(K, \theta_i, 1 - l_i)}{F_e(K, \theta_{i+1}, 1 - \hat{l}_{i+1,i})}, \tag{11}$$

yields the central intratemporal optimality condition for informal time:

$$\frac{v'(l_i)}{u'(c_i)F_e(K, \theta_i, 1 - l_i)} = \frac{1}{\rho} - \frac{\mu_{i+1}}{\rho \pi_i} \cdot \left[\frac{v'(l_i)}{u'(c_i)F_e(K, \theta_i, 1 - l_i)} - \frac{v'(\hat{l}_{i+1,i})}{u'(c_i)F_e(K, \theta_{i+1}, 1 - \hat{l}_{i+1,i})} \right]. \tag{12}$$

(The bracketed term now includes the $1/u'(c_i)$ factor required for dimensional consistency with the left-hand side.)

The left-hand side of (12) is the citizen's marginal rate of substitution between informal time and formal consumption, scaled by the marginal rate of transformation. At first-best (no incentive constraints, no political-economy frictions), this ratio equals one. The right-hand side decomposes the deviation from first-best into two components: $1/\rho$ captures the corruption wedge (uniform scaling), and the second term captures the standard Mirrleesian informational distortion, scaled by $1/\rho$. For the highest productivity type ($i = N$), $\mu_{N+1} = 0$ by convention, so the labor wedge reduces to the pure corruption distortion $1/\rho$ (a modified "no distortion at the top" result). The full economic implication is analyzed in Section 6; the derivation is in Appendix C.

5 Two Political Wedges on Public Capital

The novel theoretical content of the model is the characterization of two distinct political-economy wedges on the return to public capital.

5.1 Envelope Condition

Differentiating the maximized Bellman equation (6) with respect to current capital K and applying the envelope theorem (eliminating indirect effects through the optimized policy) yields

$$J_K(V,K) = \rho\left[\sum_i \pi_i F_K(K,\theta_i,1-l_i) + (1-\delta)\right] - \sum_i \xi_i \frac{\partial V^{\text{rev}}(K,\kappa)}{\partial K} - \phi(1-\delta). \tag{13}$$

(The detailed derivation, including cancellation of incentive-constraint terms involving $\partial \hat{l}_{i,j}/\partial K$, is in Appendix E.) From the first-order condition for next-period capital K', $\beta J_K(\bar{W},K') = 1$, so the marginal continuation value of capital is normalized to $1/\beta$ in steady state.

5.2 Effective Marginal Return

Substituting (13) into the next-period capital condition and rearranging, the effective marginal return to public capital that the contract requires for a given K' satisfies

$$\sum_i \pi_i F_K(K^*,\theta_i,1-l_i) + (1-\delta) = \frac{1}{\beta} + \frac{\sum_i \xi_i \partial V^{\text{rev}}/\partial K}{\rho} + \frac{\phi(1-\delta)}{\rho}. \tag{14}$$

Revolution tax (RT). The revolution tax arises because higher current capital raises citizens' outside option $V^{\text{rev}}(K,\kappa)$, tightening their participation constraints. The multiplier $\xi_i \geq 0$ measures the shadow value of relaxing (PC) for type i; the sum $\sum_i \xi_i \partial V^{\text{rev}}/\partial K$ is the aggregate marginal cost of higher capital through this channel. Dividing by ρ converts to resource units. The revolution tax is type-specific in origin but aggregate in the envelope; it operates only when (PC) binds for some type and is dampened by the enforcement parameter κ through $\partial V^{\text{rev}}/\partial K$.

Expropriation tax (ET). The expropriation tax arises because higher future capital K' tightens the government's no-deviation constraint $S + \beta J(\bar{W},K') \geq J^{\text{dev}}(K')$ tomorrow. The multiplier ϕ measures the shadow value of relaxing (GIC); $(1-\delta)$ is the marginal value of K' in the deviation payoff. Dividing by ρ converts to current-period units. The expropriation tax operates whenever (GIC) binds and is independent of the citizen-side participation structure. Because we evaluate J^{dev} at the installed K', the tax falls directly on the return to public investment.

Distinctness of the two wedges. The two wedges are conceptually and mathematically distinct: RT operates through the citizens' participation multipliers and is type-dependent in origin; ET operates through the government's no-deviation multiplier and is purely aggregate. They cannot be reduced to a single composite wedge by a state-variable transformation, because they have different signs and different determinants in comparative statics: RT vanishes as $\kappa \to \infty$ (perfect enforcement makes the revolution threat irrelevant) but ET does not; ET vanishes if the deviation payoff becomes flat in capital but RT does not. The interaction of the two wedges—in particular, that capital accumulation can tighten both constraints simultaneously—is, to our knowledge, new to the literature (subject to the net effect on (PC) being resolved numerically).

6 The Labor Wedge and Approximate Separation

We now examine the implications of (12) for labor distortion and the separation result of Acemoglu et al. (2010).

6.1 Decomposition

Define the labor distortion as the deviation of the marginal-rate-of-substitution-to-marginal-rate-of-transformation ratio from its first-best value of one:

$$D_i \equiv 1 - \frac{v'(l_i)}{u'(c_i)F_e(K, \theta_i, 1 - l_i)}. \tag{15}$$

Substituting the wedge formula (12) and using $\rho = 1 + \phi$ yields the additive decomposition

$$D_i = \frac{\phi}{\rho}\bigg|_{\text{aggregate corruption distortion}} + \frac{\mu_{i+1}}{\rho\pi_i} \cdot B_i\bigg|_{\text{Mirrlees informational distortion (scaled by } 1/\rho)}, \tag{16}$$

where $B_i \equiv \frac{v'(l_i)}{F_e(K,\theta_i,1-l_i)} - \frac{v'(\hat{l}_{i+1,i})}{F_e(K,\theta_{i+1},1-\hat{l}_{i+1,i})}$.

Both terms are non-negative under Assumption 1: $\phi/\rho \in [0, 1)$ is monotonically increasing in ϕ, and $B_i \geq 0$ with strict inequality whenever $\theta_{i+1} > \theta_i$. The first term is uniform across types and reflects the inflation of the resource shadow price by the government's no-deviation constraint. The second term is the standard Mirrleesian informational distortion from Mirrlees (1971), scaled by $1/\rho$. For the top type ($i = N$), $\mu_{N+1} = 0$ and $D_N = \phi/\rho$ exactly (pure corruption wedge, modified no-distortion-at-the-top).

6.2 Relation to AGT Separation

Acemoglu et al. (2010) prove that in their environment—dynamic Mirrlees taxation with a self-interested politician disciplined by replacement—political-economy distortions remain aggregate while per-type Mirrleesian wedges are unaffected. Translated to our notation, their separation result would correspond to the labor distortion taking the form $D_i^{\text{AGT}} = \Phi^{\text{agg}} + (\mu_{i+1}/\pi_i)B_i$, with the political-economy term Φ^{agg} entering additively and not interacting with the Mirrlees term.

The decomposition (16) is approximately—but not exactly—of this form. The first term ϕ/ρ is uniform across types and aggregate in spirit, consistent with AGT separation. However, the multiplier $1/\rho$ on the Mirrlees term means that the corruption wedge enters both components, and the equilibrium incentive multiplier μ_{i+1} itself depends on ϕ through general-equilibrium feedback. Strict separation therefore fails.

[Approximate Separation] The labor distortion D_i decomposes as in (16). The aggregate corruption distortion ϕ/ρ is the AGT-uniform component; the Mirrlees informational distortion $(\mu_{i+1}/(\rho\pi_i))B_i$ is the type-specific component. Strict separation fails because $\rho = 1 + \phi$ enters both terms and μ_{i+1} responds endogenously to ϕ. Numerical experiments across the plausible parameter space (detailed in Appendix H) show that the aggregate term dominates: D_i is increasing in ϕ for almost all parameter combinations, with the equilibrium response of μ_{i+1} approximately constant or weakly decreasing. The aggregate component ϕ/ρ accounts for between 63% and 94% of the total distortion at $\rho = 2$ (Table 1).

The numerical analysis underlying Lemma 1 is implemented in the companion computational paper and summarized in Table 1 (full details and replication code in Appendix H). Across 72 parameter combinations (varying $\sigma, \gamma, \alpha, \beta, \delta, \tau, \kappa, \psi$, and number of types $N = 3, 5, 10$), the labor distortion for the low-productivity type is strictly increasing in ρ (equivalently in ϕ). The indirect channel through μ_{i+1} is small and ambiguous in sign, confirming that the uniform scaling effect of corruption is the dominant force.

7 Equilibrium and Comparative Statics

[Stationary Recursive Equilibrium] A stationary recursive equilibrium is a value function J, policy functions, and a pair (V^*, K^*) such that the policy functions solve the Bellman equation (6), $\bar{W}(V^*, K^*) = V^*$, $K'(V^*, K^*) = K^*$, and all constraints are satisfied with complementary slackness.

At any stationary equilibrium, the effective return to capital (14) pins down K^* given the multipliers $\{\xi_i, \phi\}$. Because both the revolution tax and the expropriation tax are non-negative, $K^* \leq K^{\mathrm{FB}}$, with strict inequality whenever the corresponding constraints bind.

Comparative statics (conjectures). We do not derive sharp closed-form comparative statics with respect to the deep parameters (ϕ, ψ, κ) for two reasons. First, the model is sufficiently rich that several primitives affect K^* through multiple channels with offsetting signs. Second, the multiplier ϕ is endogenous, so "$\partial K^*/\partial\phi$" is not a well-defined comparative static. We summarize the directions that follow from the envelope condition (13), holding constant the equilibrium response of multipliers, as conjectures to be verified numerically in Appendix H:

1. Tighter (GIC) at the steady state $\Rightarrow$ lower K^*. Holding the participation structure fixed, an increase in the equilibrium ϕ raises the expropriation tax and lowers K^*.

2. Stronger enforcement ($\kappa \uparrow$) $\Rightarrow$ slacker (PC) $\Rightarrow$ higher K^*. Higher κ reduces $V^{\mathrm{rev}}(K, \kappa)$ and weakens the revolution tax (subject to the net effect on $\underline{V}(K, \kappa)$ being resolved numerically).

3. The effect of informal-sector strength ($\psi \uparrow$) on K^* is ambiguous. Larger ψ raises citizens' inside-contract utility, slackening (PC) and reducing the revolution tax through the participation channel. However, larger ψ also raises V^{eff}, potentially raising the outside option V^{rev} and tightening (PC). The net effect is determined numerically.

8 Conclusion

We have developed a recursive dynamic contracting model in which corruption emerges as the equilibrium outcome of two-sided physical limited commitment between a self-interested government and heterogeneous citizens with private information about their productivity. Three theoretical results stand out: the characterization of two distinct political-economy wedges on the return to public capital (the revolution tax and the expropriation tax), the approximate—but not strict—separation of corruption and Mirrleesian distortions in the per-type labor wedge, and the survival of the inverse Euler equation (in expectation) despite the political-economy frictions.

The framework is parsimonious enough to map to observable cross-country variation in corruption indices, informal-sector shares, and public-investment ratios, and rich enough to nest standard dynamic Mirrlees taxation and limited-commitment risk-sharing as special cases. We view the two-wedge characterization on public capital as the paper's main theoretical contribution. The analysis of approximate separation in the labor wedge is a more modest but novel result: it clarifies that the AGT separation result is robust to several substantial enrichments of the political-economy environment, in the sense that the uniform-scaling component remains the dominant force on labor distortion.

Full quantitative implementation, calibration to developing-economy data, and counterfactual policy experiments (e.g., the effect of raising state capacity κ on steady-state public capital and formalization) are deferred to a companion paper. The computational appendix (Appendix H) provides the replication code, parameter grid, and all numerical results underlying Lemma 1 and the comparative-statics conjectures.

A Proof of Existence (Proposition 1)

[Full proof to be included; strengthened with explicit bound on the marginal value of relaxing (GIC) to restore Blackwell contraction; alternative dual-recursive formulation noted.]

B Verification of Single Crossing (Assumption 1)

[Numerical check for baseline parametrization.]

C Full Lagrangian and Derivations of FOCs (4)–(9)

[Includes corrected scaling in labor wedge and top-type case.]

D Proof of Inverse Euler (Proposition 2)

[Explicit cancellation argument with scope conditions.]

E Envelope Condition and Derivation of Political Wedges (10)–(11)

[Full algebra.]

F Non-Convexity, Concavity, and Computational Strategy

[Discussion of Cole-Kubler issues and why dual or lottery methods suffice for quantitative work.]

G Relation to Standard Benchmarks

[Shows that when private information vanishes and revolution threat vanishes, ET collapses to the standard Thomas-Worrall/Aguiar-Amador underinvestment wedge.]

H Computational Implementation, Replication Code, and Numerical Results

[Full Python/Julia solver (value-function iteration + NLP policy recovery + steady-state finder), 72-parameter grid, Table 1, figures (labor-distortion decomposition, RT/ET magnitudes vs. κ and ψ, state-space plot with breakdown boundary), robustness checks, and calibration sketch. Code available at `https://github.com/Bonorinoa/Corruption_Contracts`.]

References